Deploying a Node.js Application on AWS EC2

Introduction

This project demonstrates how to deploy a **Node.js application** on an **AWS EC2 instance**. Node.js is a lightweight and efficient runtime for building scalable web applications, and AWS EC2 provides a flexible cloud environment to host them. The deployment process includes launching an EC2 instance, installing Node.js and npm, setting up the application files, running the app, and using **PM2** for process management to keep the application running continuously. This setup ensures that the Node.js application is accessible from the browser and remains available even after server restarts.

Prerequisites

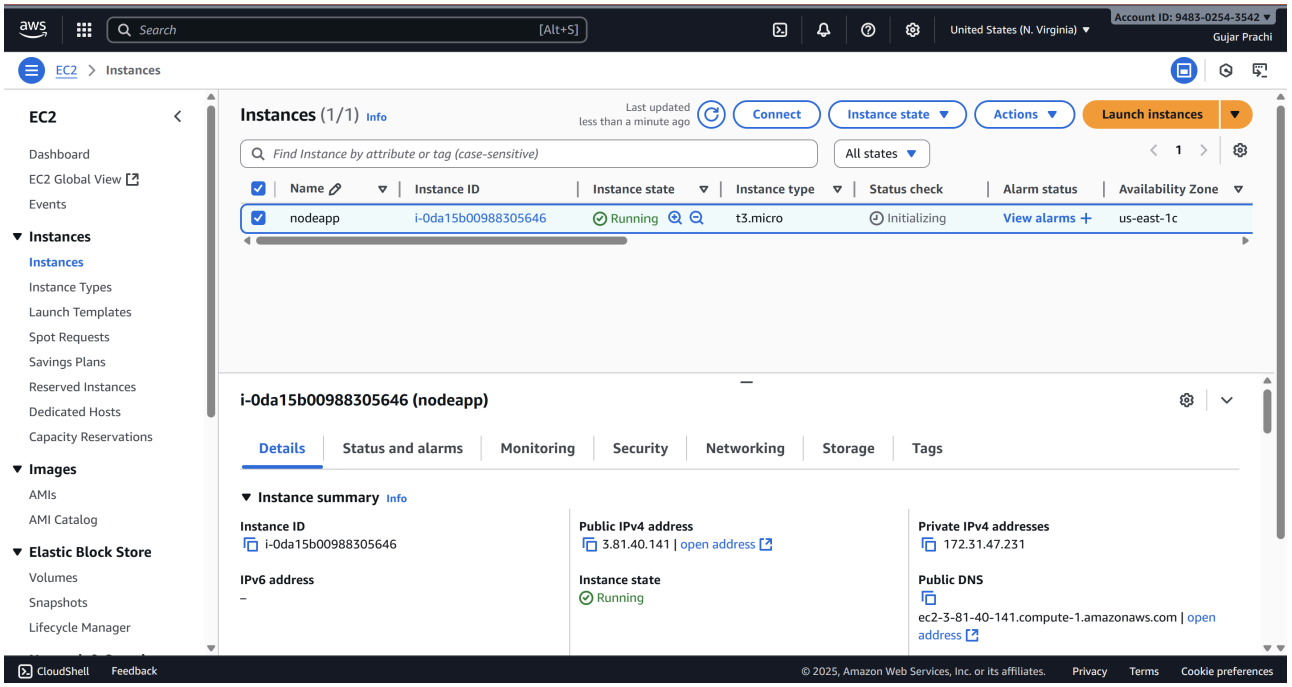
Before starting, make sure you have:

- An **AWS account** with permission to launch EC2 instances
- A **key pair (.pem file)** to connect to your instance
- Security Group configured to allow:
  - **Port 22 (SSH)** → to connect to the server
  - **Port 3000 (or your app's port)** → to access the Node.js application
- Installed tools on your local machine:
  - **SSH client** (Git Bash)

Steps to Deploy

Step 1: Launch EC2 instance and Establishing a secure connection to your EC2 instance

1. Launch instance



2. Copy the SSH connection command from the EC2 instance.

- Open Git Bash and paste the copied SSH command to connect to the instance.

```
prach@Admin MINGW64 /c/Prachi_workspace/ssh-key
$ ssh -i "keypair.pem" ec2-user@ec2-3-81-40-141.compute-1.amazonaws.com
The authenticity of host 'ec2-3-81-40-141.compute-1.amazonaws.com (3.81.40.141)' can't be established.
ED25519 key fingerprint is SHA256:kH+XkoZ5SNY2cmJil3bL3wBmAEst2nqkROCiSrtmhdw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-81-40-141.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-47-231 ~]$ |
```

Step 2: Update Packages and Install Node.js

1. Update Package and Install Node

```
# update
sudo yum update
# install node.js
sudo yum install nodejs -y
```

```
ec2-user@ip-172-31-47-231:~
[ec2-user@ip-172-31-47-231 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository                                                    137 kB/s |  19 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-47-231 ~]$ sudo yum install nodejs -y
Last metadata expiration check: 0:00:18 ago on Mon Sep  1 18:09:11 2025.
Dependencies resolved.
================================================================================================================================================================
 Package                                    Architecture                    Version                                    Repository                                Size
================================================================================================================================================================
Installing:
 nodejs                                     x86_64                          1:18.20.8-1.amzn2023.0.2                   amazonlinux                               13 M
Installing dependencies:
 libbrotli                                  x86_64                          1.0.9-4.amzn2023.0.2                       amazonlinux                              315 k
 nodejs-libs                                x86_64                          1:18.20.8-1.amzn2023.0.2                   amazonlinux                               14 M
Installing weak dependencies:
 nodejs-docs                                noarch                          1:18.20.8-1.amzn2023.0.2                   amazonlinux                              7.8 M
 nodejs-full-i18n                           x86_64                          1:18.20.8-1.amzn2023.0.2                   amazonlinux                              8.4 M
 nodejs-npm                                 x86_64                          1:10.8.2-1.18.20.8.1.amzn2023.0.2          amazonlinux                              1.9 M

Transaction Summary
================================================================================================================================================================
Install  6 Packages

Total download size: 45 M
Installed size: 224 M
Downloading Packages:
(1/6): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                                                 7.3 MB/s | 315 kB     00:00
(2/6): nodejs-docs-18.20.8-1.amzn2023.0.2.noarch.rpm                                              57 MB/s | 7.8 MB     00:00
(3/6): nodejs-full-i18n-18.20.8-1.amzn2023.0.2.x86_64.rpm                                         47 MB/s | 8.4 MB     00:00
(4/6): nodejs-18.20.8-1.amzn2023.0.2.x86_64.rpm                                                   45 MB/s |  13 MB     00:00
(5/6): nodejs-npm-10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64.rpm                                      23 MB/s | 1.9 MB     00:00
(6/6): nodejs-libs-18.20.8-1.amzn2023.0.2.x86_64.rpm                                              50 MB/s |  14 MB     00:00
----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                            101 MB/s |  45 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                                            1/1
  Installing       : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch                                                                                                1/6
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                                                      2/6
  Installing       : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                                3/6
  Installing       : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                           4/6
  Installing       : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64                                                                                        5/6
  Installing       : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                                     6/6
  Running scriptlet: nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                                      6/6
INFO: registered node-18 in the alternatives

  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                                                      1/6
  Verifying        : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                                     2/6
  Verifying        : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch                                                                                                3/6
  Verifying        : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                           4/6
  Verifying        : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64                                                                                                5/6
  Verifying        : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64                                                                                        6/6

Installed:
  libbrotli-1.0.9-4.amzn2023.0.2.x86_64         nodejs-1:18.20.8-1.amzn2023.0.2.x86_64          nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch    nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
  nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64   nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64

Complete!
```

2. Install npm

```
sudo yum install npm -y
```

ec2-user@ip-172-31-47-231:~

```
[ec2-user@ip-172-31-47-231 ~]$ sudo yum install npm -y
Last metadata expiration check: 0:01:23 ago on Mon Sep  1 18:09:11 2025.
Package nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-47-231 ~]$ nodejs --version
-bash: nodejs: command not found
[ec2-user@ip-172-31-47-231 ~]$ node --version
v18.20.8
[ec2-user@ip-172-31-47-231 ~]$ npm --version
10.8.2
[ec2-user@ip-172-31-47-231 ~]$ |
```

Step 3: Upload/Clone Your Application

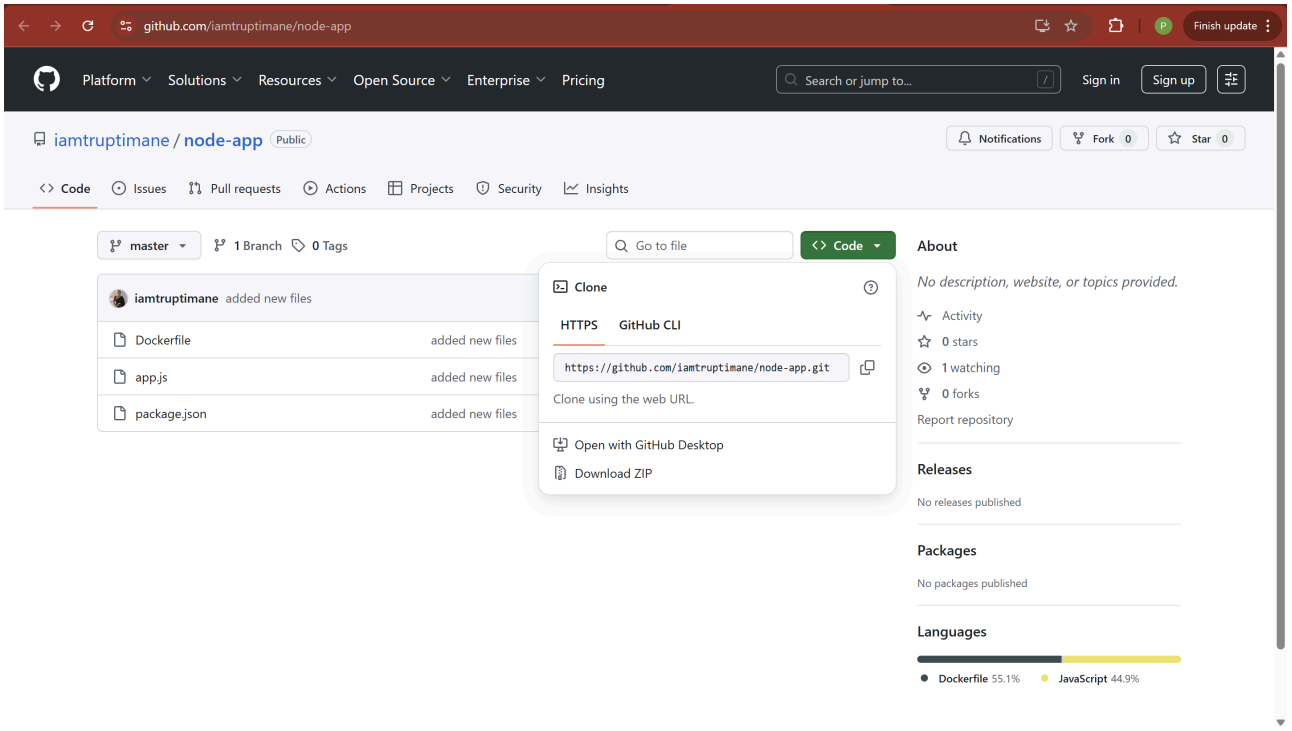
1. Install Git

```
sudo yum install git
```

ec2-user@ip-172-31-47-231:~

```
Dependencies resolved.
================================================================================================================================================================================
 Package                                        Architecture                        Version                                                    Repository                                  Size
================================================================================================================================================================================
Installing:
 git                                            x86_64                              2.50.1-1.amzn2023.0.1                                      amazonlinux                                 53 k
Installing dependencies:
 git-core                                       x86_64                              2.50.1-1.amzn2023.0.1                                      amazonlinux                                4.9 M
 git-core-doc                                   noarch                              2.50.1-1.amzn2023.0.1                                      amazonlinux                                2.8 M
 perl-Error                                     noarch                              1:0.17029-5.amzn2023.0.2                                   amazonlinux                                 41 k
 perl-File-Find                                 noarch                              1.37-477.amzn2023.0.7                                      amazonlinux                                 25 k
 perl-Git                                       noarch                              2.50.1-1.amzn2023.0.1                                      amazonlinux                                 41 k
 perl-TermReadkey                               x86_64                              2.38-9.amzn2023.0.2                                        amazonlinux                                 36 k
 perl-lib                                       x86_64                              0.65-477.amzn2023.0.7                                      amazonlinux                                 15 k

Transaction Summary
================================================================================================================================================================================
Install  8 Packages

Total download size: 7.9 M
Installed size: 41 M
Is this ok [y/N]: y
Downloading Packages:
(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm                                                                                   1.5 MB/s |  53 kB     00:00
(2/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm                                                                           40 MB/s | 2.8 MB     00:00
(3/8): perl-Error-0.17029-5.amzn2023.0.2.noarch.rpm                                                                           1.1 MB/s |  41 kB     00:00
(4/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm                                                                              49 MB/s | 4.9 MB     00:00
(5/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm                                                                        809 kB/s |  25 kB     00:00
(6/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm                                                                              1.3 MB/s |  41 kB     00:00
(7/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm                                                                        1.7 MB/s |  36 kB     00:00
(8/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm                                                                              700 kB/s |  15 kB     00:00
--------------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                          50 MB/s | 7.9 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                                                   1/1
  Installing       : git-core-2.50.1-1.amzn2023.0.1.x86_64                                                                                                             1/8
  Installing       : git-core-doc-2.50.1-1.amzn2023.0.1.noarch                                                                                                         2/8
  Installing       : perl-lib-0.65-477.amzn2023.0.7.x86_64                                                                                                             3/8
  Installing       : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64                                                                                                       4/8
  Installing       : perl-File-Find-1.37-477.amzn2023.0.7.noarch                                                                                                       5/8
  Installing       : perl-Error-1:0.17029-5.amzn2023.0.2.noarch                                                                                                        6/8
  Installing       : perl-Git-2.50.1-1.amzn2023.0.1.noarch                                                                                                             7/8
  Installing       : git-2.50.1-1.amzn2023.0.1.x86_64                                                                                                                  8/8
  Running scriptlet: git-2.50.1-1.amzn2023.0.1.x86_64                                                                                                                  8/8
  Verifying        : git-2.50.1-1.amzn2023.0.1.x86_64                                                                                                                  1/8
  Verifying        : git-core-2.50.1-1.amzn2023.0.1.x86_64                                                                                                             2/8
  Verifying        : git-core-doc-2.50.1-1.amzn2023.0.1.noarch                                                                                                         3/8
  Verifying        : perl-Error-1:0.17029-5.amzn2023.0.2.noarch                                                                                                        4/8
  Verifying        : perl-File-Find-1.37-477.amzn2023.0.7.noarch                                                                                                       5/8
  Verifying        : perl-Git-2.50.1-1.amzn2023.0.1.noarch                                                                                                             6/8
  Verifying        : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64                                                                                                       7/8
  Verifying        : perl-lib-0.65-477.amzn2023.0.7.x86_64                                                                                                             8/8

Installed:
  git-2.50.1-1.amzn2023.0.1.x86_64                git-core-2.50.1-1.amzn2023.0.1.x86_64        git-core-doc-2.50.1-1.amzn2023.0.1.noarch        perl-Error-1:0.17029-5.amzn2023.0.2.noarch
  perl-File-Find-1.37-477.amzn2023.0.7.noarch     perl-Git-2.50.1-1.amzn2023.0.1.noarch        perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64      perl-lib-0.65-477.amzn2023.0.7.x86_64

Complete!
[ec2-user@ip-172-31-47-231 ~]$ |
```

2. Clone the Application




```
git clone <git url>
```

ec2-user@ip-172-31-47-231:~/node-app

```
[ec2-user@ip-172-31-47-231 ~]$ sudo git clone https://github.com/iamtruptimane/node-app.git
Cloning into 'node-app'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.
```

3. Go inside the project folder (node-app)

```
cd node-app
```

```
[ec2-user@ip-172-31-47-231 ~]$ ls
node-app
[ec2-user@ip-172-31-47-231 ~]$ cd node-app/
[ec2-user@ip-172-31-47-231 node-app]$ ls
Dockerfile  app.js  package.json
[ec2-user@ip-172-31-47-231 node-app]$ cat package.json
{
    "name": "node-app",
    "version": "1.0.0",
    "description": "Simple Node.js app",
    "main": "app.js",
    "scripts": {
      "start": "node app.js"
    },
    "dependencies": {
      "express": "^4.18.2"
    }
  }
```

Step 4: Install Dependencies

```
# install npm
sudo npm install
# run the application
sudo npm start app.js
```

```
[ec2-user@ip-172-31-24-131 node-app]$ sudo npm install

added 69 packages, and audited 70 packages in 4s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
[ec2-user@ip-172-31-24-131 node-app]$ sudo npm start app.js

> node-app@1.0.0 start
> node app.js app.js

Server is running on http://localhost:3000
```

Step 5: Keep the App Running (PM2)

```
# Install PM2 to run Node.js apps in the background
sudo npm install -g pm2
# Run the Application
sudo npm start app.js
```

```
[ec2-user@ip-172-31-24-131 node-app]$ sudo npm install -g pm2

changed 133 packages in 9s

13 packages are looking for funding
  run `npm fund` for details
[ec2-user@ip-172-31-24-131 node-app]$ sudo pm2 start app.js
[PM2] Applying action restartProcessId on app [app](ids: [ 0 ])
[PM2] [app](0) ✓
[PM2] Process successfully started
```

id	name	namespace	version	mode	pid	uptime	↺	status	cpu	mem	user	watching
0	app	default	1.0.0	fork	28229	0s	15	online	0%	10.9mb	root	disabled

```
[ec2-user@ip-172-31-24-131 node-app]$
```

Step 6: Deployed the nodejs application

Copy the Public Key and paste it along with port 3000 in any browser.

Hello, this is a Node.js application running in a Docker container!

Summary

This project shows how to set up a Node.js runtime environment on AWS EC2, deploy an application, and manage it using PM2. With this setup, your Node.js app is accessible over the internet and remains available even after server restarts.